MILITARY SPECIFICATION

RIOT CONTROL AGENT, CR

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of riot control agent, dibenz (b,f)-1,4-oxazepine, hereinafter referred to as CR.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forma part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

FSC 1365

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 1500.42, Vol 38, No. 187, dated Sep 27, 1973 - Consumer Product Safety Commission Regulations
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "16 CFR 1500.42" and "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water
E324 - Relative Initial and Final Melting Points, and the Melting Range of Industrial Organic Chemicals

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedences

3. REQUIREMENTS

3.1 Volatile matter. CR shall contain no more than 1.0 percent by weight volatile matter when tested as specified in 4.2.3.1.

3.2 Melting range. CR shall have a melting range between 67° and 74°C when tested as specified in 4.2.3.2.

3.3 Purity. CR shall have a purity of no less than 95 percent by weight when tested as specified in 4.2.3.3.

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3.4 Eye irritancy. CR shall not produce persistent eye irritancy when tested as specified in 4.2.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the CR produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling and inspection procedure.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordante with MIL-STD-105 using an AQL of 2.5 percent defective. The sample unit shall be one filled unit or shipping container, as applicable, ready for ship ment. Sample unit and shipping-containers shall be examined for the following defects:

(a) Net contents not as specified
(b) Container not as specified
(c) Container damaged or leaking
(d) Exterior of container contaminated with CR
(e) Marking incorrect, missing, or illegible
(f) Palletization not as specified (when required)

4.2.2.2 For test (see 6.7). Three specimens of approximately 25 grams (g) each shall be removed from different parts of the lot of CR offered for acceptance and placed in separate clean, dry, closed containers labeled to identify the lot represented. Each specimen shall be tested as specified in 4.2.3.1, 4.2.3.2, and 4.2.3.3. Failure of any test by any specimen shall be cause for rejection of the lot represented. For the eye irritancy test, approximately 5 g shall be taken from each of the three 25-g specimens and blended to form a composite. This composite shall be tested as specified in 4.2.3.4. Failure of this test by the composite shall be cause for rejection of the lot represented

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4.2.3 Tests (see 6.7). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.3.1 Volatile matter. Weigh to the nearest milligram (mg) approximately 5 g of specimen into a glass-stoppered weighing bottle approximately 5 centimeters (cm) in diameter. Weigh the bottle and contents to the nearest milligram, remove the stopper, and place the stopper and uncovered weighing bottle in a vacuum desiccator over freshly activated silica gel. Close the desiccator, evacuate the air, and maintain the vacuum (10 millimeters of mercury or less) for 24 hours. Release the vacuum slowly, passing the incoming air through a suitable drying train. Stopper the bottle and reweigh. Retain the dried specimen for use in 4.2.3.2. Calculate the percent by weight volatile matter as follows:

$$\text{volatile matter} = \frac{100(A - B)}{W}$$

where: A = Weight in grams of stoppered bottle and contents before drying,
B = Weight in grams of stoppered bottle and contents after drying, and
W = Weight in grams of specimen.

4.2.3.2 Melting range. Determine the melting range of the dried material retained from 4.2.3.1 in accordance with ASTM E324.

4.2.3.3 Purity. Determine the purity of the specimen by the ultraviolet (UV) spectrophotometer method specified in 4.2.3.3.1 or by the gas-liquid chromatography method specified in 4.2.3.3.2.

4.2.3.3.1 UV spectrophotometer.

(a) Standard solutions. Weigh to the nearest 0.1 mg approximately 0.2 g of recrystallized CR (see 6.5) having a melting range of 71° to 74°C when tested as specified in 4.2.3.2. Quantitatively transfer the CR to a 1-liter volumetric flask. Dissolve the CR and dilute to the volume mark with 1N hydrochloric acid. Pipet aliquots of 2, 4, 6, 8, and 10 milliliters (ml) of the solution into separate 100-ml volumetric flasks. Dilute each aliquot to the volume mark with 1N hydrochloric acid and mix thoroughly.

(b) Specimen solution. Prepare a specimen solution by substituting the specimen (as received and without drying) for the recrystallized CR in procedure (a).

(c) Procedure. Using a UV spectrophotometer suitable for measuring absorbance at 315 nanometers, cells of 1 cm path length, and 1N hydrochloric acid in the reference beam, measure the absorbance of the standard solutions and the specimen solution at the peak occurring between 315 and 320

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nanometers. Plot the absorbance of each diluted standard solution against its concentration in milligrams per liter using rectangular coordinate paper. Plot the standard curve using the method of least squares as the line determining procedure. Calculate the concentration of each diluted solution as follows:

Concentration of diluted solution in milligrams per liter = 10 AB

where: A = Milliliters in transfered stock solution aliquot and
B = Concentration of stock solution in milligrams per milliliter.

Calculate the percent by weight CR as follows:

$$\text{Percent CR} = \frac{100AB}{CW}$$

where: A = Absorbance of specimen solution,
B = Weight in grams of recrystallized CR,
C = Absorbance of standard solution, and
W = Weight in grams of specimen.

4.2.3.3.2 Gas-liquid chromatography.

(a) Apparatus. Use a gas-liquid chromatography with isothermal or temperature programming. Recommended chromatographic conditions for a Perkin-Elmer Model 900 chromatography are shown in table I. These conditions may be modified as necessary when other chromatography are used.

TABLE I. Chromatographic conditions

Characteristic	Condition
Detector	Thermal conductivity
Bridge current, milliamps	150
Attenuation	8X (as needed)
Carrier gas	Helium
Carrier gas flow rate, ml per minute	90
Detector temperature, °C on column injection chamber temperature, °C	180
Programmed column temperature	6°C/min from 65° to 200°C
Column material	"Pyrex" glass
Column dimensions	5.5 feet by 1/4 inch OD
Column support	60/80 mesh "gas chrom Q"
Column coating	10 percent "QF-1"

(b) Determination of response factor. Prepare a 10-ml ethanol solution containing approximately 0.5 g of CR of known purity (see 6.6) and 0.5 g of an internal standard of known purity. (1,10-Dibromodecane has been found satisfactory for use as an internal standard; other materials may be used.)

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Weigh the CR and the internal standard to the nearest 0.1 mg into a 10-ml volumetric flask. Dissolve and dilute to the volume mark with ethanol. Saturate the chromatography column three times with 20-microliter portions of the solution. Inject a fourth 20-microliter portion of the solution and measure the resulting areas for the CR and the internal standard. Calculate the response factor for CR, corrected for purity, as follows:

$$\text{Response factor} = \frac{AB}{CD}$$

where: A = Area per unit weight of the CR of known purity,
B = Percent purity of the internal standard,
C = Area per unit weight of the internal standard, and
D = Percent purity of the CR.

(c) Procedure. Weigh to the nearest milligram approximately 0.5 g each of the specimen and the internal standard into a 10-ml volumetric flask. Dissolve and dilute to the volume mark with ethanol. Saturate the chromatography column three times with 20-microliter portions of the solution. Inject a fourth 20-microliter portion of the solution and measure the resulting areas. (Using conditions in table I, 1,10-dibromodecane will elute in approximately 19 minutes at 173°C and the specimen will elute in approximately 21 minutes at 185°C). Calculate the percent by weight CR as follows:

$$\text{Percent CR} = \frac{EFG}{HRW}$$

where: E = Area produced by the specimen solution,
F = Weight in grams of the internal standard,
G = Percent purity" of the internal standard,
H = Area produced by the internal standard solution,
R = Response factor calculated in (b), and
W = Weight in grams of specimen.

4.2.3.4 Eye irritancy. Determine eye irritancy of the specimen in accordance with 16 CFR 1500.42, Test for Eye Irritants, with the following modifications:

(a) For paragraph (a)(1) sentence 8, 0.1 ml of a solution of 1 percent by weight specimen in propylene glycol/water (80/20) is to be used.

(b) For paragraph (b)(1), substitute the following: the animal shall be considered as exhibiting a positive reaction if the test substance produces more than a mild reddening of the eyelid margins and slight injection of the blood vessels of the conjunctival and nictitating membrane.

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5. PACKAGING

5.1 Industrial packing. Quantities of CR as specified (see 6.2) shall be packed in a manner to maintain CR purity. It shall be packed to meet carrier acceptance and safe delivery to destination at lowest rates in compliance with requirements of carrier rules and regulations applicable to the mode of transportation (see 6.8). Exterior of containers shall be free of CR contamination.

5.2 Marking. Containers shall, as a minimum, be legibly marked to show the following information:

(a) Noun nomenclature cited in the contract or purchase order
(b) National Stock Number (or part number when National Stock Number is not given) cited in the contract or purchase order
(c) Government contract or purchase order number
(d) Quantity
(e) Lot or batch number
(f) Date of manufacture and packaging
(g) Contractor's name
(h) Appropriate address markings on exterior containers
(i) Precautionary and hazard markings required by MIL-STD-129, Department of Transportation regulations, and any other applicable regulations

5.3 Unitization. When specified in the contract or order (see 6.2), shipping containers shall be palletized to meet carrier acceptance and safe delivery to destination at lowest rates in compliance with requirements of carrier rules and regulations applicable to the mode of transportation.

6. NOTES

6.1 Intended use. CR is intended for use in the manufacture of Riot Control Agent, CR Solution.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Net contents per container (see 5.1).
(c) If palletization is required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufacture-d by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off “to the nearest unit” in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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6.5 Recrystallization of CR. The following procedure has been found satisfactory for recrystallizing CR for use in the purity determination: Dissolve the CR using heat in the minimum amount of ethanol necessary. Add sufficient water to initiate precipitation of the CR plus a slight excess. Filter through a coarse filter paper by gravity, discarding the initial precipitate. Add additional water to the filtrate. Filter and allow the CR on the filter paper to dry. Redissolve the CR from the second filtration in ethanol and evaporate to dryness. Determine the melting range. If the melting range does not fall within a 71° to 74°C range, recrystallize the CR.

6.6 Standard CR. CR of known purity for use in 4.2.3.3.2(b) may be obtained from Commander, US Army Armament Research and Development Command, Product Assurance Directorate, Chemical Systems Division, Aberdeen Proving Ground, MD 21010. Alternatively, the contractor may prepare his own standard CR and obtain verification of its purity from Chemical Systems Division.

6.7 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations and procedures must be followed in the sampling and testing of this material.

6.8 Packaging precaution. Styrofoam beads used as filler material cause static charge buildup making handling of CR difficult and should not be used.

6.9 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-R-1135, dated 9 October 1975.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A071

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-R-51446(EA) Riot Control Agent, CR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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MILITARY SPECIFICATION

RIOT CONTROL AGENT, CR

This amendment forms a part of Military Specification MIL-R-51446(EA), dated 28 April 1982, and is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 8

Add the following paragraph after 6.9:

6.10 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement QSTAG 495. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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Preparing activity:

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MIL-R-51446(EA)
NOTICE 2
8 November 1995

SUPERSEDING
NOTICE 1
1 June 1989

MILITARY SPECIFICATION

RIOT CONTROL AGENT, CR

This notice should be filed in front of MIL-R-51446(EA), dated 28 April 1982, with Amendment 1, dated 30 August 1982.

MIL-R-51446(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

AMSC N/A

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